

Customer Support RAG Chatbot using Langflow, ChromaDB & OpenAI GPT-4.1

Project Overview

This project implements a Retrieval-Augmented Generation (RAG) chatbot using Langflow, ChromaDB, OpenAI text-embedding-3-large and GPT-4.1. A Customer FAQ PDF is ingested, split into chunks, embedded, stored in ChromaDB, and queried through semantic search to answer customer questions.

Workflow

Read File → Split Text (1000 chunk, 200 overlap) → OpenAI Embeddings → ChromaDB (ingestion).

Chat Input → ChromaDB Search → Parser → Prompt Template → GPT-4.1 → Chat Output.

Technologies

Langflow, OpenAI GPT-4.1, OpenAI Embeddings, ChromaDB, Python.

Features

Document ingestion, semantic search, context-aware answers, low-code workflow, local vector database.

Sample Questions

Shipment times, order tracking, returns, payment methods, bulk discounts, customer support.

Future Enhancements

Multiple documents, conversation memory, source citations, Docker deployment, Streamlit frontend.